The conditional mean minimizes squared-error risk

Context Prior slide: conditional risk and centered norm expansion

$$m = E[z|y], \quad C = \mathrm{Cov}[z|y], \quad r = z - m, \quad a \in \mathbb{R}^d, \quad E[\|z\|^2|y] < \infty$$

$$\begin{aligned} R(a|y) &= E[\|z-a\|^2|y] \\ &= E[\|r\|^2|y] + 2E[r^T(m-a)|y] + \|m-a\|^2 \end{aligned}$$

$$E[r|y] = E[z|y] - m = 0 \quad \rightarrow \quad E[r^T(m-a)|y] = E[r|y]^T(m-a) = 0$$

$$E[\|r\|^2|y] = \sum_{j=1}^{d} E[r_j^2|y] = \sum_{j=1}^{d} C_{jj} = \mathrm{tr}\ C$$

$$R(a|y) = \mathrm{tr}\ C + \|a-m\|^2$$

$$\arg\min_{a\in\mathbb{R}^d} R(a|y) = \{m\}, \quad \min_{a\in\mathbb{R}^d} R(a|y) = \mathrm{tr}\ C$$

$$p(z|y) = \mathcal{N}\left(z|\mu_{z|y}, \Sigma_{z|y}\right), \quad m = \mu_{z|y}, \quad C = \Sigma_{z|y}, \quad \hat{z} = \mu_{z|y}$$

- The conditional mean uniquely minimizes squared-error risk; Gaussianity is unnecessary.
- The minimum risk is the covariance trace; displacement from the mean adds squared distance.